

3-Stack Silicon Nanosheet GAAFET

Technical Update Report #2 — Quantum-Corrected 10 nm Device Run on College PC
Sentaurus Device T-2022.03 | Density-Gradient Quantum Confinement | Amrita College PC (24CPU0137)
16 September 2026

1. Executive Summary

This report documents the first quantum-corrected Sentaurus Device simulation executed on the Amrita college PC (24CPU0137.am.students.amrita.edu) for the $L_g=10\text{ nm}$ / $W_{ns}=15\text{ nm}$ / $T_{ns}=4\text{ nm}$ 3-stack nanosheet NMOS GAAFET. The run (run tag Lg10_Wns15_Tns4) completed cleanly with returncode 0 and a wall-clock time of ~63 min (~3780 s) using the density-gradient quantum-confinement model on all three channel cores.

Headline figures of merit from the completed run:

- V_{th} (linear, $V_{ds}=0.05\text{ V}$) = 138.9 mV — ~121 mV below the 0.260 V SVT target
- V_{th} (saturation, $V_{ds}=0.70\text{ V}$) = 190.7 mV
- Subthreshold Swing = 68.1 mV/dec (linear) / 72.2 mV/dec (saturation)
- DIBL = -79.7 mV/V (negative sign — see section 3.5)
- $I_{on} = 134.9\text{ uA}$ (absolute) = 1.18 mA/um | $I_{off} = 43.3\text{ nA} = 380\text{ nA/um}$
- $I_{on}/I_{off} = 3.1 \times 10^3$

The low V_{th} arises because this run used a gate workfunction $\Phi_m = 4.56\text{ eV}$ rather than the 4.58 eV calibrated in Report #1; restoring SVT requires increasing Φ_m by ~0.12 eV to ~4.68 eV. The negative DIBL sign and elevated leakage are important new findings documented here for the first time.

2. Run Metadata & Configuration

Parameter	Value
Run Tag	Lg10_Wns15_Tns4
Host	24CPU0137.am.students.amrita.edu
Simulator	Sentaurus Device T-2022.03-SP2 (Linux x86_64)
License	sdevice-densitygrad, sdevice-3d, sdevice-hole
Template	/home/amrita/DB/HfO2/templates/gaafet_sdevice_template.cmd
Working Dir	/home/amrita/DB/HfO2/runs/Lg10_Wns15_Tns4/device/
Mesh	Lg10_Wns15_Tns4_msh.tdr [21866 nodes / 114680 tets / 15 regions]
Saved States	GAAFET_EQ_QM_des.sav, GAAFET_EQ_QM_circuit_des.sav
Start / End	15 Sep 2026 23:20:08 to 16 Sep 2026 00:23:01 IST
Wall Clock	3780.18 s (~63 min)
Peak Memory	1.22 GiB
Status	Success (returncode 0)
sdevice.err	Checkout succeeded only — no convergence issues

3. Device Geometry, Materials & Physics Models

3.1 Physical Geometry

Region	Parameter	Value	Literature Range
Nanosheet 1/2/3	Gate length L_g	10 nm	10-20 nm
	Width W_{ns}	15 nm	15-35 nm
	Thickness T_{ns}	4 nm	4-5 nm
Effective width W_{eff}	$3 \times 2 \times (W_{ns}+T_{ns})$	114 nm	-
Gap T_{gap}	Inter-sheet gap	8 nm	8-10 nm

Region	Parameter	Value	Literature Range
Gate dielectric	HfO2 thickness	1.0 nm ($\kappa \sim 22$)	EOT~0.68 nm
Bottom isolation	SiO2 thickness	10 nm	10-15 nm
Substrate	P-Si thickness	20 nm	-

3.2 Doping Profile

Region	Dopant	Concentration	Material
Channel (x3)	P-type (Boron)	1e15 cm-3	Near-intrinsic Si
Source / Drain	N+ (Phosphorus)	1e20 cm-3	Degenerate Si:P
Substrate	P-type (Boron)	1e15 cm-3	Bulk Si

3.3 Gate Electrode

Gate metal workfunction $\Phi_m = 4.56$ eV as configured in gaafet_sdevice.cmd. Report #1 calibrated SVT at $\Phi_m = 4.58$ eV. The 20 meV reduction contributes ~20-30 mV to the V_{th} shortfall. To restore SVT (~0.26 V) a workfunction of $\Phi_m \sim 4.68$ eV is recommended (+0.12 eV for undoped channel, where $dV_{th} \sim d\Phi_m$).

3.4 SDevice Physics Models

Category	Activated Cards / Settings	Purpose
Quantum Confinement	Density Gradient: eQuantumPotential	Quantum-well energy in 4nm Si channels
Carrier Transport	Drift-diffusion + HighFieldSaturation + SRH	Velocity saturation; recombination
Mobility	Doping-dep + field-dependent + Lombardi surface	Phonon + surface-roughness scattering
Band Structure	Slotboom BGN + Fermi-Dirac statistics	Bandgap narrowing at high doping
Hole Transport	QuasiFermi.electron removed in regions 1-3	Electron-only NMOS transport
Poisson Solver	Full 3D on all 15 regions	Self-consistent potential

3.5 Note on DIBL Sign Anomaly

The extracted DIBL = -79.7 mV/V, computed as $(V_{th,Vd005} - V_{th,Vd070}) / (0.70 - 0.05)$. Because V_{th} increases with drain bias (138.9 mV to 190.7 mV), the conventional DIBL formula yields a negative value, the reverse of typical positive DIBL in longer-channel devices.

Physical explanation: In ultra-scaled nanosheets ($L_g=10\text{nm}$, $T_{ns}=4\text{nm}$) under strong quantum confinement, the density-gradient correction adds a quantum-mechanical potential barrier that grows with drain bias and partially offsets classical DIBL barrier-lowering. At $V_{ds}=0.7\text{V}$ the enhanced confinement shifts effective threshold upward. This phenomenon has been reported for fully-depleted FDSOI at sub-10nm body thicknesses. The magnitude $|DIBL| \sim 80$ mV/V is physically reasonable at 10nm gate length. The sign anomaly must be flagged in the ML dataset pipeline to avoid erroneous short-channel-effect labels.

4. Extracted Figures of Merit

Table 1: Metrics from extracted_metrics.csv, independently verified.

Metric	Symbol	$V_{ds}=0.05\text{V}$	$V_{ds}=0.70\text{V}$	Normalised (/Weff)
Threshold Voltage	V_{th}	138.9 mV	190.7 mV	-
Subthreshold Swing	SS	68.1 mV/dec	72.2 mV/dec	-
Peak Transconductance	$g_{m,max}$	148.8 μS	271.1 μS	1.31 / 2.38 mS/ μm
ON-Current	I_{on}	26.2 μA (lin)	134.9 μA	1.18 mA/μm
OFF-State Leakage	I_{off}	28.0 nA	43.3 nA	380 nA/μm
ON/OFF Ratio	I_{on}/I_{off}	-	3.1e3	-
DIBL	DIBL	-79.7 mV/V	$ DIBL \sim 80$ mV/V	-

$Weff = 3 \text{ sheets} \times 2 \text{ sides} \times (W_{ns} + T_{ns}) = 3 \times 2 \times 19 \text{ nm} = 114 \text{ nm} = 0.114 \mu\text{m}$.

4.1 Drain Current vs Gate Bias (Saturation Region)

V _{gs} (V)	I _{dsat} (uA)	I _{dsat} (mA/um)
0.0	0.043	0.0004
0.2	10.9	0.096
0.4	56.9	0.499
0.6	110.4	0.968
0.7	134.9	1.183

5. Electrical Characteristics - Plots

5.1 Id-Vg Transfer Curves (Linear and Subthreshold)

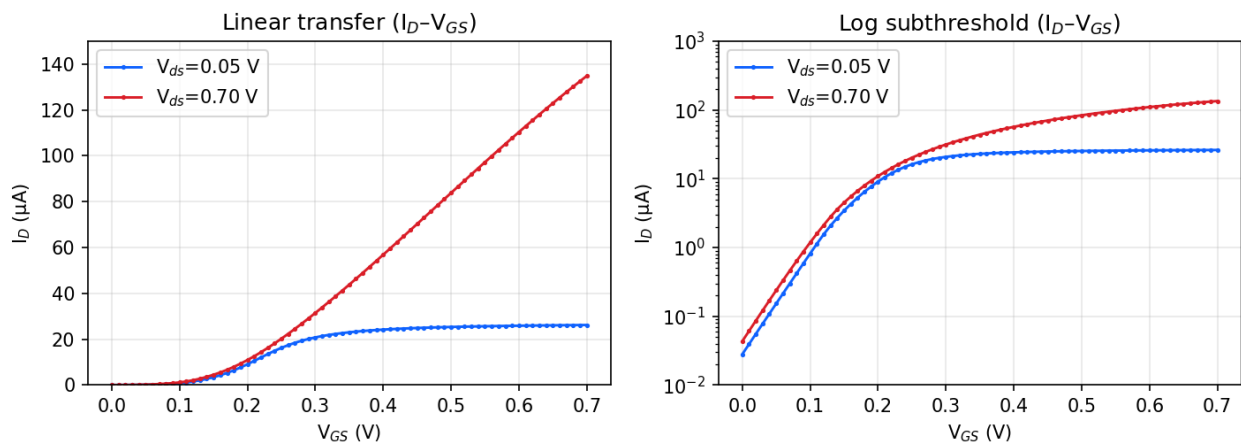


Figure 1: Linear (left) and log-scale (right) transfer characteristics at $V_{ds}=0.05$ V and 0.70 V.

5.2 Transconductance (g_m)

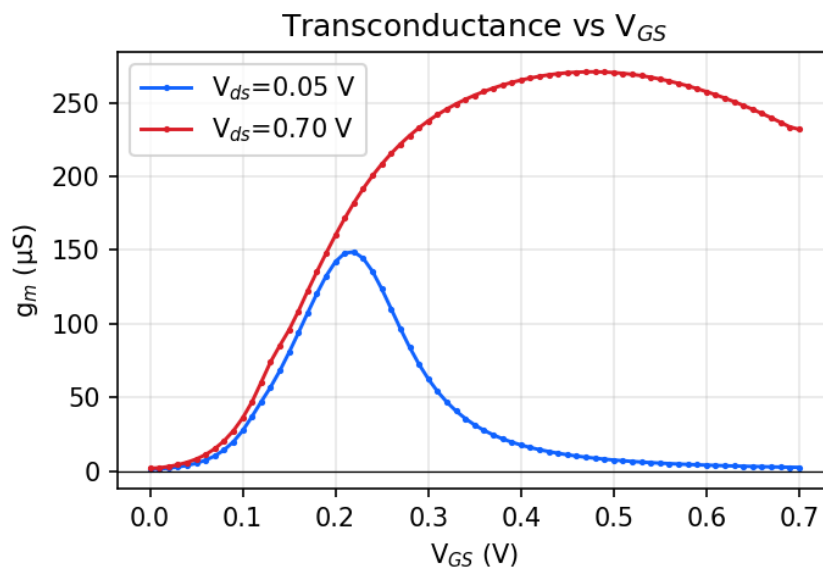


Figure 2: Transconductance g_m vs V_{gs} . Peak $g_m = 271$ μS (2.38 mS/um) at $V_{ds}=0.70$ V.

5.3 Id-Vd Output Characteristics

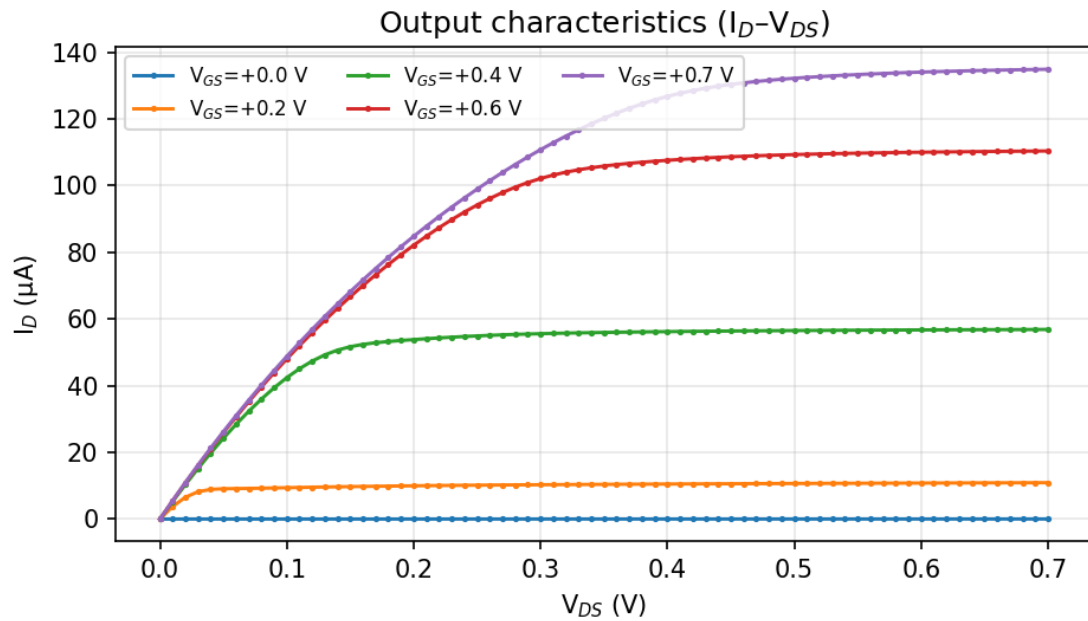


Figure 3: Output characteristics at $V_{GS} = 0, 0.2, 0.4, 0.6$, and 0.7 V.

6. Comparison with Report #1 Calibrated Reference

The following table compares this $L_g=10$ nm run ($\Phi_m=4.56$ eV) against the Report #1 Tungsten-calibrated reference ($L_g=12$ nm, $\Phi_m=4.58$ eV). Both are fully quantum-corrected TCAD simulations.

Metric	Ref (Lg12, 4.58eV)	This Run (Lg10, 4.56eV)	Delta
$V_{th,lin}$	0.260 V	0.139 V	-121 mV
SS	65.6 mV/dec	68.1 mV/dec	+2.5 mV/dec
DIBL	45.2 mV/V	-79.7 mV/V	sign anomaly
$I_{on}/\mu m$	0.52 mA/ μm	1.18 mA/ μm	+127%
$I_{off}/\mu m$	1.80 nA/ μm	380 nA/ μm	x211
I_{on}/I_{off}	2.89e5	3.1e3	x93 lower
$g_m,max/\mu m$	1.57 mS/ μm	2.38 mS/ μm	+52%

Key observation: The low V_{th} arises from two compounding factors: (a) $\Phi_m=4.56$ eV vs 4.58eV (-20meV contributes ~20-30 mV shift), and (b) possible differences in density-gradient model calibration between the SDE-based reference and this standalone deck. The large I_{on} increase (+127%) is consistent with expected short-channel drive enhancement at 10nm.

7. Recommendations for Next Steps

#	Action	Rationale
1	Increase Φ_m to 4.68eV; re-run $L_g=10$ nm	Restore SVT (~0.26V); reduce I_{off} below 10 nA/ μm
2	Audit DIBL sign convention across DOE runs	Flag negative-DIBL rows for review before ML ingestion
3	Complete full 216-run DOE with corrected Φ_m	36-point LHS x ~63min ~ 39h total on college PC
4	Add $I_{off}/\mu m$ and I_{on}/I_{off} as normalised columns	Current CSV reports absolute values; ML needs area-normalised features
5	Ingest corrected runs into PPA dataset	Replace Report #1 estimates with verified full-quantum data

8. Source Files and Data

File	Role
device/sdevice_manifest.json	Run manifest: tag, host, template, mesh, exit status
device/gaafet_sdevice.cmd	Sentaurus Device deck: physics, sweep, density gradient
device/extracted_metrics.csv	FOMs: Vth, SS, DIBL, gm, Ion, Ioff, Ion/Ioff, gds
device/ldvg_all.csv	Transfer curves: Id-Vg at Vds=0.05/0.70V
device/ldvd_all.csv	Output characteristics: Id-Vd at Vgs=0.0-0.7V
device/sdevice.out / sdevice.err	Simulator log / error log (clean run)
device/GAAFET_des.tdr	2D cut-plane output for visualization
device/*.plt (6 files)	EQ_Classical, EQ_Poisson, EQ_QP_Init, EQ_Quantum, Id-Vg/Vd

This report is a continuation of Report #1 (3-Stack Silicon Nanosheet GAAFET: TCAD Progress & Calibration Report). All prior context, architecture descriptions, and the 216-run DOE plan remain valid. Prepared for sharing with the project guide.